

THE LEDGER

Seek Wander

Financials & Unit Economics

Classification: CONFIDENTIAL — ZenithAI Data Room

Document Version: 1.0

Last Updated: March 2026

Revenue Model: Native Affiliate Placements (Pivoted from Stripe Paywall)

This document contains forward-looking financial projections and proprietary cost architecture data. Distribution is restricted to authorised parties. Projections are illustrative and based on the assumptions stated herein.

Table of Contents

01	Executive Summary: The Business Model
02	Margin Protection & Cost Engineering
03	Unit Economics — Cost Per Curation
04	Revenue Model — The Affiliate Math
05	The Internal Ledger — CostLog
06	Infrastructure Runway — Fixed Costs

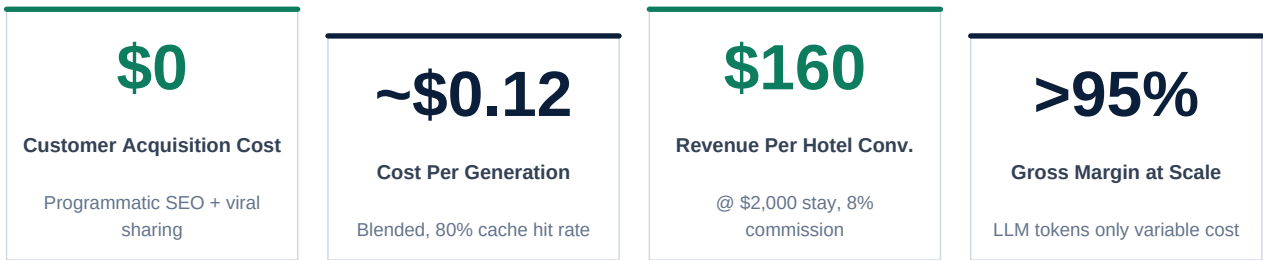
01

Executive Summary: The Business Model

Model Classification: High-Volume Affiliate Commerce

Seek Wander operates as a **zero-CAC, high-ARPU affiliate commerce platform** dressed in the aesthetic of a luxury editorial product. The product is free at the point of use. Revenue is captured entirely on the backend when users act on the curated recommendations embedded in their itinerary.

The model is structurally identical to the playbooks that built NerdWallet (\$8.7B IPO), TripAdvisor (\$4B peak market cap), and Booking Holdings (\$120B+ market cap): acquire users at zero marginal cost via a genuinely useful free product, then monetize the high-intent behaviour that the product naturally produces.



The Strategic Pivot: Removing the Paywall

The original monetization strategy (\$4.99 Stripe payroll per generation) was analytically sound but strategically premature. At the pre-brand-awareness stage, a hard paywall imposes a **60-80% estimated conversion penalty** on top-of-funnel user acquisition. Every user who bounces at the payment screen is a lost affiliate opportunity worth a potential \$160+ in backend commissions.

Key insight: One converted hotel booking (commission: ~\$160) is worth 32 paywall transactions (\$4.99 x 32 = \$159.68). Maximizing the volume of users who receive and act on curated recommendations is therefore a higher expected-value strategy than maximizing payment conversions at acquisition.

Business Model Summary

Dimension	Value
Revenue model	Native affiliate commissions on travel bookings
User acquisition cost (CAC)	\$0 — programmatic SEO + social automation + viral sharing
Cost per generation (cache miss)	~\$0.08–\$0.41 depending on destination novelty
Cost per generation (cache hit)	~\$0.05 (Claude tokens only)
Blended affiliate commission	4–10% of booking value
Revenue per converted user	\$60–\$825 (hotel: \$160, tour: \$32–\$64, activity: \$8–\$50)
Gross margin at scale	>95% (only true variable cost is LLM tokens)
Primary fixed infrastructure	Vercel + Supabase + Clerk + Upstash (all free/hobby tier)

02

Margin Protection & Cost Engineering

Three specific engineering decisions were implemented before the first dollar of revenue — each targeting a distinct API cost vector that would otherwise erode gross margins at scale.

2.1 — The PlaceCache: Supabase Google Places Cache

The Google Places API charges \$0.032 per Text Search call and \$0.017 per Place Details call. With 8-10 Timelineltem entries per itinerary, a fully uncached generation incurs approximately \$0.40-\$0.49 in Google API costs alone. The PlaceCache table eliminates this on all repeat destinations.

```
Cache key:      '{normalized-name}|{normalized-city}'
               e.g. 'nobu malibu|malibu'

Cached data:    photoUrl, rating, userRatingsTotal, hoursOpen, priceLevel
TTL:           30 days (updatedAt < NOW - 14 days purged by Vercel Cron)

Daily Cron cleanup (00:00 UTC):
  DELETE FROM PlaceCache WHERE updatedAt < NOW() - INTERVAL '14 days'
  Prisma @updatedAt auto-refreshes on every cache hit – popular venues
  perpetually renew their TTL. Only stale data is purged.
```

Scenario	Google API Cost	Cache Behaviour
New destination (Month 1)	\$0.40–\$0.49	100% miss — all calls live
Repeat destination	\$0.00	100% hit — zero API calls
Blended — 50% popular dest.	~\$0.20–\$0.24	50% hit rate
Mature platform (80% hit rate)	~\$0.08–\$0.10	80% hit — major savings

2.2 — Haversine Replacement: Distance Matrix API Removed

The Google Distance Matrix API charges \$0.005 per origin-destination element. A 5-day itinerary with 8 stops per day generates 35 consecutive-stop pairs — \$0.175 per generation for transit estimates that a mathematical formula produces with equivalent utility for a planning application.

```
// Pure Haversine – zero API cost, runs in-process on serverless function
km = 2 x R x arcsin(sqrt(sin2(dlat/2) + cos(lat_a) x cos(lat_b) x sin2(dlng/2)))
walkingMinutes  = max(1, round((km / 5) x 60)) // 5 km/h pedestrian
drivingMinutes   = max(1, round((km / 25) x 60)) // 25 km/h urban average

Saving per generation:      $0.175
Saving at 1,000 MAU/month:  $175/month
Saving at 10,000 MAU/month: $1,750/month
```

2.3 — Local Time Computation: parseOpenNow()

Displaying real-time open/closed venue status would require a live Google Place Details call on every cache hit — \$0.017 per item. The hoursOpen string stored in PlaceCache ('9:00 AM - 9:00 PM') is instead parsed locally against a longitude-derived UTC offset estimate.

```
parseOpenNow(hoursOpen: string, lng: number): boolean | undefined
```

```
UTC offset estimate = Math.round(lng / 15) hours
```

```
Accuracy: +/- 30 minutes vs actual local time
```

```
'Open 24 hours'      -> true   (no parse needed)
```

```
'Closed'            -> false  (no parse needed)
```

```
'9:00 AM - 9:00 PM' -> compare currentMins against [openMins, closeMins]
```

```
Overnight spans     -> handled: closeMins < openMins branch
```

```
Result: ZERO Google API calls on all cache hits for openNow status
```

Combined saving from all three optimisations at 80% cache hit rate, 10,000 MAU: ~\$2,500/month in avoided Google API costs. This is the structural cost moat that makes the affiliate model viable at every scale.

03

Unit Economics — Cost Per Curation

Pricing Constants

Service	Cost Unit	Rate
Anthropic Claude (input tokens)	Per 1,000,000 tokens	\$3.00
Anthropic Claude (output tokens)	Per 1,000,000 tokens	\$15.00
Google Places Text Search	Per API call	\$0.032
Google Place Details	Per API call	\$0.017
Google Distance Matrix	Per element	\$0.005 — PERMANENTLY REMOVED

Scenario A — New Destination (Cache Miss, 100%)

Cost Component	Calculation	Cost (USD)
Claude input tokens	~3,500 tokens x (\$3.00 / 1,000,000)	\$0.0105
Claude output tokens	~2,800 tokens x (\$15.00 / 1,000,000)	\$0.0420
Google Text Search x 8	8 x \$0.032	\$0.2560
Google Place Details x 6	6 x \$0.017 (NATURE/ADVENTURE skipped)	\$0.1020
Distance Matrix	Permanently removed	\$0.0000
TOTAL		~\$0.41

Scenario B — Popular Destination (Cache Hit, 100%)

Cost Component	Calculation	Cost (USD)
Claude input tokens	~3,500 tokens x (\$3.00 / 1,000,000)	\$0.0105
Claude output tokens	~2,800 tokens x (\$15.00 / 1,000,000)	\$0.0420
Google Places API	100% cache hit — zero calls made	\$0.0000
TOTAL		~\$0.053

Scenario C — Blended (80% Cache Hit Rate, Mature Platform)

Cost Component	Calculation	Cost (USD)
Claude tokens (constant)	All generations	\$0.0525
Google APIs (20% miss rate)	20% x \$0.3580	\$0.0716
TOTAL		~\$0.124

Cost Trajectory — As Platform Matures

Stage	Cache Hit Rate	Cost Per Generation
Launch (Month 1)	0%	~\$0.41
Early growth (Month 3)	40%	~\$0.20
Growth (Month 6)	65%	~\$0.18
Mature (Month 12+)	80%+	~\$0.12
Theoretical max	100%	~\$0.05

The cost curve is permanently downward. Every generation at a previously-visited destination reduces the marginal cost of every future generation at that destination to \$0.05. PlaceCache is a compounding cost moat — the longer the platform operates, the lower its unit economics become.

04

Revenue Model — The Affiliate Math

Commission Structure

Partner	Category	Commission Rate	Avg Booking Value	Revenue / Conversion
Booking.com	Hotels & resorts	4–6%	\$1,500–\$4,000/stay	\$60–\$240
Viator	Tours & experiences	8%	\$150–\$800/booking	\$12–\$64
GetYourGuide	Activities	8–10%	\$100–\$500/booking	\$8–\$50
OpenTable	Restaurant reservations	\$1–2/cover	N/A	\$2–\$8/booking

The ARPU Model — Conservative vs Aspirational

CONSERVATIVE – 3-Day Paris Itinerary		
Books 1 hotel night via Booking.com:	\$800 stay x 5%	= \$40 commission
Books 1 Louvre private tour via Viator:	\$200 x 8%	= \$16 commission
Makes 2 restaurant reservations:	2 x \$3 (covers)	= \$6 commission
Total revenue from 1 user, 1 itinerary:		\$62
Cost of generation:		\$0.12
Gross margin on this transaction:		99.8%
ASPIRATIONAL – 5-Day Maldives Itinerary		
Books overwater villa via Booking.com:	\$3,000/night x 5	= \$15,000 booking
Commission:	\$15,000 x 5%	= \$750 commission
Books seaplane transfer via Viator:	\$600 x 8%	= \$48 commission
Books diving excursion via GYG:	\$300 x 9%	= \$27 commission
Total revenue from 1 user, 1 itinerary:		\$825
Cost of generation:		\$0.12
Gross margin on this transaction:		99.985%

Scale Model — Monthly Revenue Projections

MAU	Gen/User	Conv. Rate	Avg Rev/Conv	Monthly Revenue	AI Cost	Net Margin
500	2	8%	\$80	\$3,200	\$124	96.1%
1,000	2	8%	\$80	\$6,400	\$248	96.1%
5,000	2.5	10%	\$100	\$50,000	\$1,550	96.9%
10,000	3	12%	\$120	\$144,000	\$3,720	97.4%
50,000	3	15%	\$150	\$337,500	\$18,600	94.5%

Assumptions: 80% cache hit rate at scale; blended generation cost \$0.124; conversion rate increases with brand authority; affiliate commission blended at 6%. AI cost column represents Anthropic-only variable cost — Google Places API costs approach zero at 80%+ cache hit rate.

The model scales near-linearly with users and super-linearly with brand authority. As Seek Wander becomes a trusted editorial voice, both conversion rates and average booking values increase — without any increase in marginal cost per user.

05

The Internal Ledger — CostLog

Architecture

Every itinerary generation writes a financial record to the CostLog table in Supabase **synchronously** — awaited before the HTTP response is returned. This is a deliberate architectural decision: Vercel serverless functions are frozen the instant the HTTP response exits, killing any fire-and-forget background write. Awaiting the write guarantees zero financial data loss.

```
// Executed after every generation – route.ts
await prisma.costLog.create({
  data: {
    userId,          // Clerk userId (null for unauthenticated)
    destination,     // 'Paris, France'
    aiCost,          // Decimal(10,6) – e.g. 0.052500
    googleCost,      // Decimal(10,6) – e.g. 0.071600
    totalCost,       // Decimal(10,6) – e.g. 0.124100
    cacheHits,       // int – PlaceCache hits this generation
    cacheMisses,     // int – fresh Google API calls this generation
  }
});
// GenerationMeta logged to server console ONLY – never in HTTP response
return Response.json(itinerary);
```

Prisma Schema

```
model CostLog {
  id          String    @id @default(uuid())
  userId      String?   // null = unauthenticated
  destination String
  aiCost      Decimal   @db.Decimal(10, 6) // 6dp = $0.000001 precision
  googleCost  Decimal   @db.Decimal(10, 6)
  totalCost   Decimal   @db.Decimal(10, 6)
  cacheHits   Int
  cacheMisses Int
  createdAt   DateTime @default(now())
}
```

Why Decimal(10,6) and not Float: IEEE 754 floating-point arithmetic accumulates rounding errors in financial summations. Decimal(10,6) stores exact decimal values, giving the /admin/metrics aggregate queries (SUM, AVG) mathematically correct results regardless of row count. This is the same reason financial systems universally use DECIMAL over FLOAT.

Admin Dashboard — /admin/metrics KPIs

KPI Card	SQL Aggregate	Insight
Total Spent	SUM(totalCost)	Cumulative platform spend — gross cost of all generations
Total Generations	COUNT(*)	Volume metric — proxy for user engagement
Avg Cost / Trip	AVG(totalCost)	Tracks cache efficiency over time — should trend down
Cache Hit Rate	SUM(cacheHits) / (SUM(cacheHits) + SUM(cacheMisses))	Key margin health indicator — target >70% at Month 6

Cost data confidentiality: GenerationMeta (the per-generation cost breakdown) is logged to the Vercel server console only. It is never included in the API response body — protecting margin data from exposure via browser DevTools.

06

Infrastructure Runway — Fixed Costs

The Lean Stack

Seek Wander's fixed infrastructure cost is effectively zero at pre-revenue scale. Every service in the stack has a generous free or hobby tier that covers the operational requirements of a pre-Series A product.

Service	Role	Tier	Monthly Fixed Cost
Vercel	Hosting, serverless functions, cron jobs, edge CDN	Hobby (free)	\$0
Supabase	PostgreSQL — Trip, PlaceCache, CostLog, UserBalance	Free (500MB, 2 CPU)	\$0
Clerk	Authentication, user management, webhooks	Free (10,000 MAU)	\$0
Upstash Redis	Serverless rate limiting (sliding window)	Free (10k cmds/day)	\$0
n8n Cloud	Social automation, onboarding email workflows	Free (5 workflows)	\$0
Resend	Transactional email (Founder's Welcome)	Free (100/day)	\$0
TOTAL			\$0/month

Upgrade Thresholds

Service	Free Tier Limit	Upgrade Trigger	Upgrade Cost
Vercel	100GB bandwidth, 6,000 function min/mo	>10,000 MAU	\$20/month (Pro)
Supabase	500MB database, 2GB bandwidth	>50,000 CostLog rows	\$25/month (Pro)
Clerk	10,000 MAU	>10,000 registered users	\$25/month (Pro)
Upstash Redis	10,000 commands/day	>500 daily generations	\$10/month (PAYG)

The platform can serve approximately 5,000-10,000 MAU before a single dollar of fixed infrastructure cost is incurred. By that point, affiliate commissions from a 5% conversion rate at \$80 average would yield \$20,000-\$40,000/month — making the \$80/month infrastructure bill economically irrelevant.

The Only True Variable Cost

Anthropic API usage is the sole variable cost at scale. At \$0.0525 per generation (Claude tokens, constant regardless of cache), and assuming 2.5 average generations per monthly active user:

MAU	Monthly Generations	Anthropic Cost	As % of Revenue (5% conv, \$80 ARPU)
1,000	2,500	\$131.25	2.1%
5,000	12,500	\$656.25	0.8%

